DevOps



LAB 02

Hands-On Git and Version Control

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BCS 7-B

Department Of Computer Science

Getting Started: Install Git

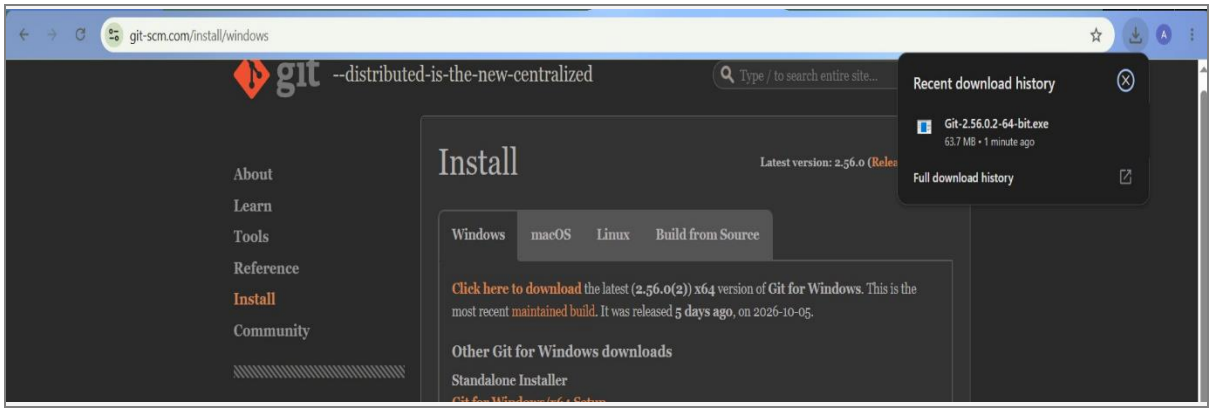


Figure: Git installed on my PC (version check).

Task 1: Create Private GitHub Repository

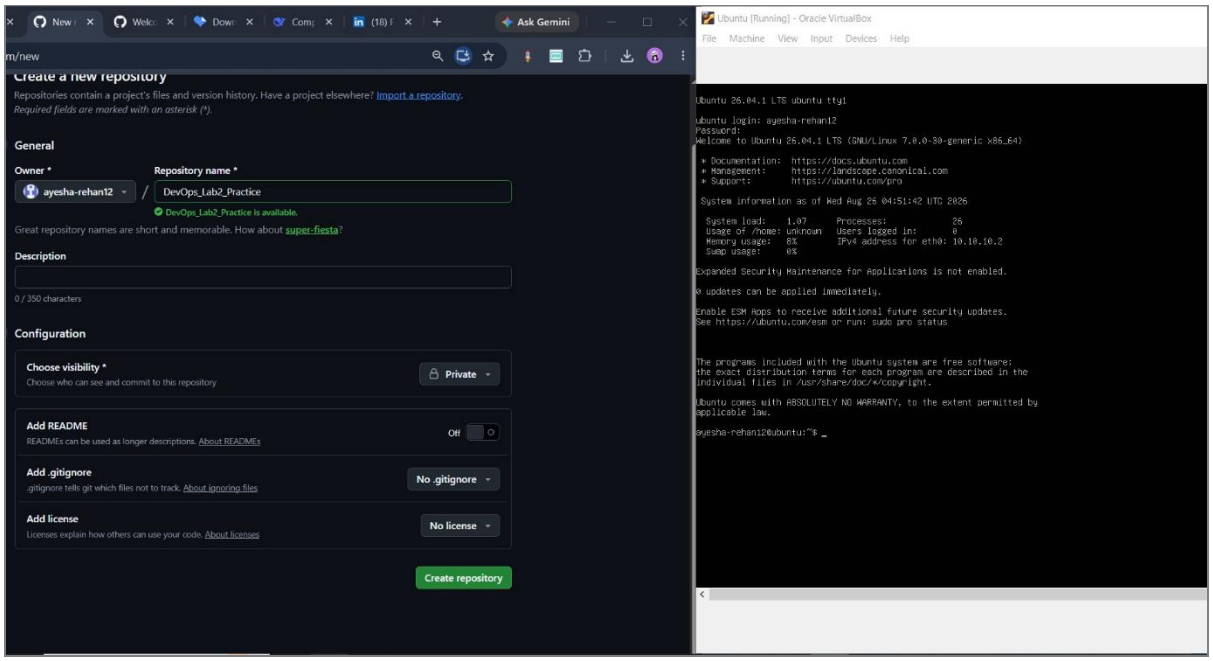


Figure: DevOps_Lab2_Practice repository is private, with my terminal prompt visible.

Task 2: Connect Repository via SSH

```
ayesha-rehan12@ubuntu:~$ ssh-keygen -t ed25519 -C "421ayesharehan@gmail.com"
Generating public/private ed25519 key pair.
Enter file in which to save the key (/home/ayesha-rehan12/.ssh/id_ed25519):
Enter passphrase for "/home/ayesha-rehan12/.ssh/id_ed25519" (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/ayesha-rehan12/.ssh/id_ed25519
Your public key has been saved in /home/ayesha-rehan12/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:cIyOHSM7TDlJ1XQl1AV1RXvSFFv1S36TbewVgBMQZ9g 421ayesharehan@gmail.com
The key's randomart image is:
+--[ED25519 256]--+
|     ...o.+O*==oo@|
|    . o o.ooE. .+=|
|     * = o    . .=+|
|    o B =       oo*|
|     = o S       =*|
|      .          o+|
|                  .|
|                   |
|                   |
+----[SHA256]-----+
ayesha-rehan12@ubuntu:~$ cat ~/.ssh/id_ed25519.pub
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIHQ9yaCnT7ncX2768sAdoogtLUED9kI6Odcmm54pewRZ 421ayesharehan@gmail.com
ayesha-rehan12@ubuntu:~$
```

Figure: SSH key generated.

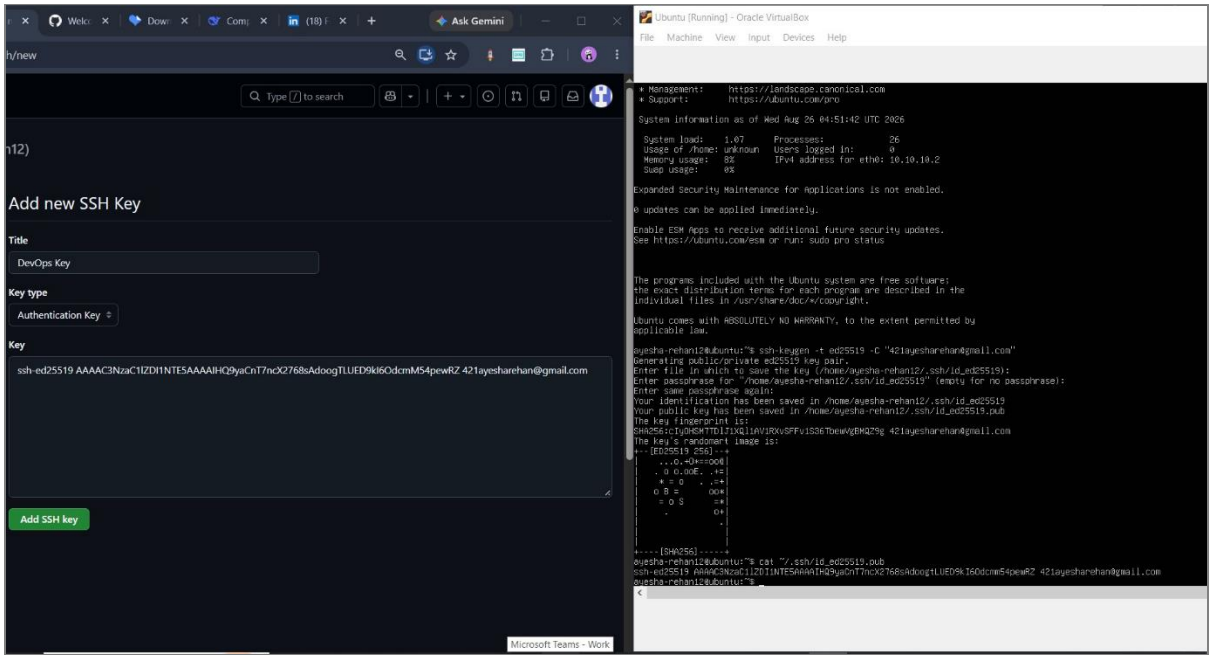


Figure: SSH public key added in GitHub Settings > SSH and GPG keys.

```
ayesha-rehan12@ubuntu:~$ ssh -T git@github.com
Hi ayesha-rehan12! You've successfully authenticated, but GitHub does not provide shell access.
ayesha-rehan12@ubuntu:~$ git clone git@github.com:ayesha-rehan12/DevOps_Lab2_Practice.git
Cloning into 'DevOps_Lab2_Practice'...
warning: You appear to have cloned an empty repository.
ayesha-rehan12@ubuntu:~$
```

Figure: Repository cloned using SSH.

Task 3: Configure Git Username and Email

```
and the repository exists.
ayesha-rehan12@ubuntu:~$ git config --global user.name "Ayesha Rehan"
ayesha-rehan12@ubuntu:~$ git config --global user.email "421ayesharehan@gmail.com"
ayesha-rehan12@ubuntu:~$
ayesha-rehan12@ubuntu:~$
ayesha-rehan12@ubuntu:~$
ayesha-rehan12@ubuntu:~$
ayesha-rehan12@ubuntu:~$
```

Figure: Git username and email configured globally.

```
ayesha-rehan12@ubuntu:~$
ayesha-rehan12@ubuntu:~$
ayesha-rehan12@ubuntu:~$ git config --list
user.name=Ayesha Rehan
user.email=421ayesharehan@gmail.com
ayesha-rehan12@ubuntu:~$
ayesha-rehan12@ubuntu:~$
ayesha-rehan12@ubuntu:~$
```

Figure: Output of git config --list verifying the configuration.

Task 4: Explore the .git Folder

```
ayesha-rehan12@ubuntu:~/DevOps_Lab2_Practice$ ls -a .git
.  ..  HEAD  config  description  hooks  info  objects  refs
ayesha-rehan12@ubuntu:~/DevOps_Lab2_Practice$
```

Figure: Contents of the .git directory shown with ls -a .git.

Task 5: Local Repository Management

```
ayesha-rehan12@ubuntu:~/DevOps_Lab2_Practice$ rm -rf .git
ayesha-rehan12@ubuntu:~/DevOps_Lab2_Practice$ git status ==> error why?
fatal: not a git repository (or any of the parent directories): .git
ayesha-rehan12@ubuntu:~/DevOps_Lab2_Practice$
```

Figure: .git folder deleted and git status showing the error.

```
ayesha-rehan12@ubuntu:~/DevOps_Lab2_Practice$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: will change to "main" in Git 3.0. To configure the initial branch name
hint: to use in all of your new repositories, which will suppress this warning,
hint: call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
hint:
hint: Disable this message with "git config set advice.defaultBranchName false"
Initialized empty Git repository in /home/ayesha-rehan12/DevOps_Lab2_Practice/.git/
ayesha-rehan12@ubuntu:~/DevOps_Lab2_Practice$
```

Figure: Local repository re-initialized with git init.

```
ayesha-rehan12@ubuntu:~/DevOps_Lab2_Practice$ echo "# Lab02 Git Practice" >> README.md
ayesha-rehan12@ubuntu:~/DevOps_Lab2_Practice$ git add README.md
ayesha-rehan12@ubuntu:~/DevOps_Lab2_Practice$ git commit -m "Initial commit"
[master (root-commit) c3e5e5d] Initial commit
 1 file changed, 1 insertion(+)
 create mode 100644 README.md
ayesha-rehan12@ubuntu:~/DevOps_Lab2_Practice$
```

Figure: README.md created and committed as Initial commit.

```
ayesha-rehan12@ubuntu:~/DevOps_Lab2_Practice$ git remote add origin git@github.com:ayesha-rehan12/DevOps_Lab2_Practice.git
ayesha-rehan12@ubuntu:~/DevOps_Lab2_Practice$ git branch -M main
ayesha-rehan12@ubuntu:~/DevOps_Lab2_Practice$ git push -u origin main
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Writing objects: 100% (3/3), 239 bytes | 239.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:ayesha-rehan12/DevOps_Lab2_Practice.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
ayesha-rehan12@ubuntu:~/DevOps_Lab2_Practice$
```

Figure: Remote origin added and main branch pushed to GitHub.

Task 6: File Status & Staging

```
ayesha-rehan12@ubuntu:~/DevOps_Lab2_Practice$ echo "Adding some test notes" >> notes.txt
ayesha-rehan12@ubuntu:~/DevOps_Lab2_Practice$ git status
On branch main
Your branch is up to date with 'origin/main'.

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        error
        notes.txt

nothing added to commit but untracked files present (use "git add" to track)
ayesha-rehan12@ubuntu:~/DevOps_Lab2_Practice$
```

Figure: git status showing notes.txt as untracked.

```
ayesha-rehan12@ubuntu:~/DevOps_Lab2_Practice$ git add notes.txt
ayesha-rehan12@ubuntu:~/DevOps_Lab2_Practice$ git commit -m "Add notes.txt"
[main 8ea7ef7] Add notes.txt
 1 file changed, 1 insertion(+)
 create mode 100644 notes.txt
ayesha-rehan12@ubuntu:~/DevOps_Lab2_Practice$
```

Figure: notes.txt staged and committed.

Task 7: Branch Creation Using GitHub GUI

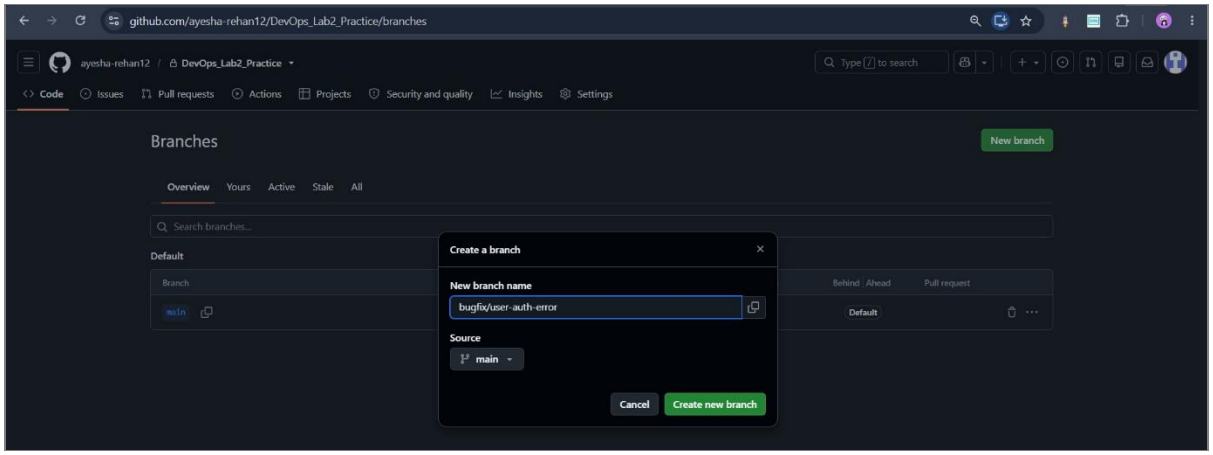


Figure: Branch bugfix/user-auth-error created using the GitHub web interface.

```
ayesha-rehan12@ubuntu:~/DevOps_Lab2_Practice$ git pull origin bugfix/user-auth-error
From github.com:ayesha-rehan12/DevOps_Lab2_Practice
 * branch            bugfix/user-auth-error -> FETCH_HEAD
 * [new branch]      bugfix/user-auth-error -> origin/bugfix/user-auth-error
Already up to date.
ayesha-rehan12@ubuntu:~/DevOps_Lab2_Practice$
```

Figure: Local branch list after pulling bugfix/user-auth-error.

Task 8: Branch Creation and Push Using Git Bash

```
ayesha-rehan12@ubuntu:~/DevOps_Lab2_Practice$ git checkout -b feature/db-connection
Switched to a new branch 'feature/db-connection'
ayesha-rehan12@ubuntu:~/DevOps_Lab2_Practice$ git push origin feature/db-connection
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 303 bytes | 303.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'feature/db-connection' on GitHub by visiting:
remote:      https://github.com/ayesha-rehan12/DevOps_Lab2_Practice/pull/new/feature/db-connection
remote:
To github.com:ayesha-rehan12/DevOps_Lab2_Practice.git
 * [new branch]      feature/db-connection -> feature/db-connection
ayesha-rehan12@ubuntu:~/DevOps_Lab2_Practice$
```

Figure: Branch feature/db-connection created locally and pushed to remote.

Task 9: Branching & Merging

```
ayesha-rehan12@ubuntu:~/DevOps_Lab2_Practice$ git checkout -b feature-1
Switched to a new branch 'feature-1'
ayesha-rehan12@ubuntu:~/DevOps_Lab2_Practice$
```

Figure: Branch feature-1 created and switched to.

```
ayesha-rehan12@ubuntu:~/DevOps_Lab2_Practice$ cat main.py
def greet(name):
    return f"Hello, {name}!"

def add_numbers(a, b):
    return a + b
ayesha-rehan12@ubuntu:~/DevOps_Lab2_Practice$ python3 -c "from main import add_numbers; print(add_numbers(2, 3))"
5
ayesha-rehan12@ubuntu:~/DevOps_Lab2_Practice$
ayesha-rehan12@ubuntu:~/DevOps_Lab2_Practice$
ayesha-rehan12@ubuntu:~/DevOps_Lab2_Practice$
ayesha-rehan12@ubuntu:~/DevOps_Lab2_Practice$ git add main.py
ayesha-rehan12@ubuntu:~/DevOps_Lab2_Practice$ git commit -m "Add new function to main.py"
[feature-1 58f172a] Add new function to main.py
 1 file changed, 5 insertions(+)
 create mode 100644 main.py
ayesha-rehan12@ubuntu:~/DevOps_Lab2_Practice$
```

Figure: New function added to main.py and committed on feature-1.

```
 create mode 100644 main.py
ayesha-rehan12@ubuntu:~/DevOps_Lab2_Practice$ git checkout main
Switched to branch 'main'
Your branch is ahead of 'origin/main' by 1 commit.
  (use "git push" to publish your local commits)
ayesha-rehan12@ubuntu:~/DevOps_Lab2_Practice$ git merge feature-1
Updating 8ea7ef7..58f172a
Fast-forward
 main.py | 5 +++++
 1 file changed, 5 insertions(+)
 create mode 100644 main.py
ayesha-rehan12@ubuntu:~/DevOps_Lab2_Practice$
```

Figure: Switched to main and merged feature-1.

```
ayesha-rehan12@ubuntu:~/DevOps_Lab2_Practice$ git push origin main
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 394 bytes | 131.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:ayesha-rehan12/DevOps_Lab2_Practice.git
   c3e5e5d..58f172a  main -> main
ayesha-rehan12@ubuntu:~/DevOps_Lab2_Practice$ git push origin feature-1
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'feature-1' on GitHub by visiting:
remote:      https://github.com/ayesha-rehan12/DevOps_Lab2_Practice/pull/new/feature-1
remote:
To github.com:ayesha-rehan12/DevOps_Lab2_Practice.git
 * [new branch]      feature-1 -> feature-1
ayesha-rehan12@ubuntu:~/DevOps_Lab2_Practice$
```

Figure: main and feature-1 branches pushed to remote.

Task 10: Pull Request and Branch Review (GitHub GUI)

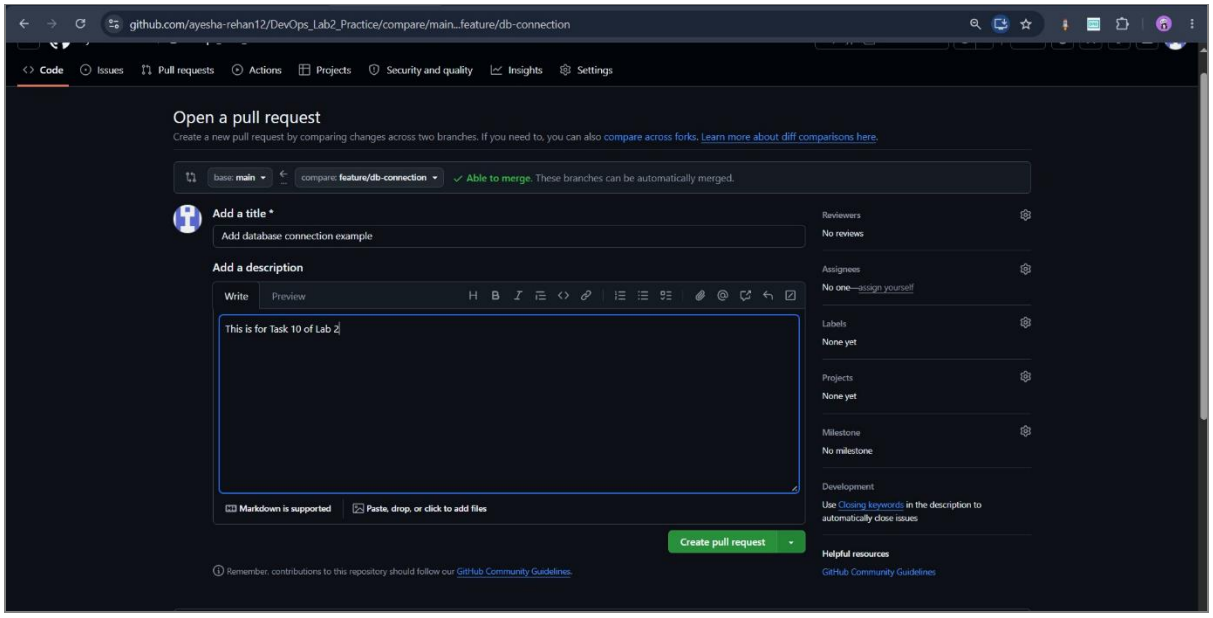


Figure: Pull Request created from feature/db-connection to main.

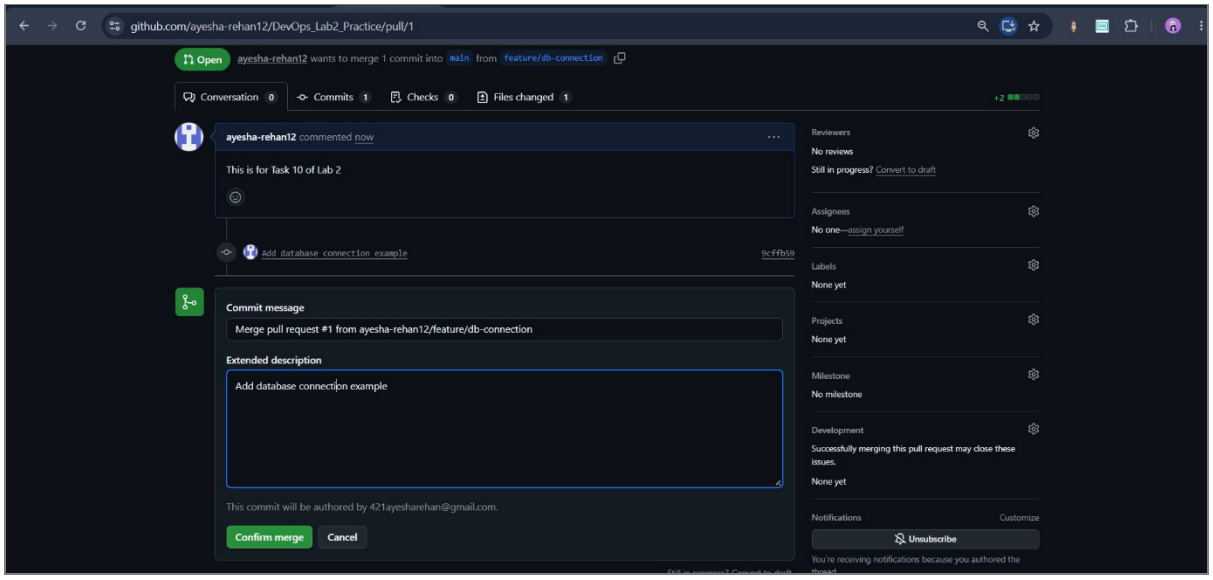


Figure: Pull Request reviewed and merged using the GitHub GUI.

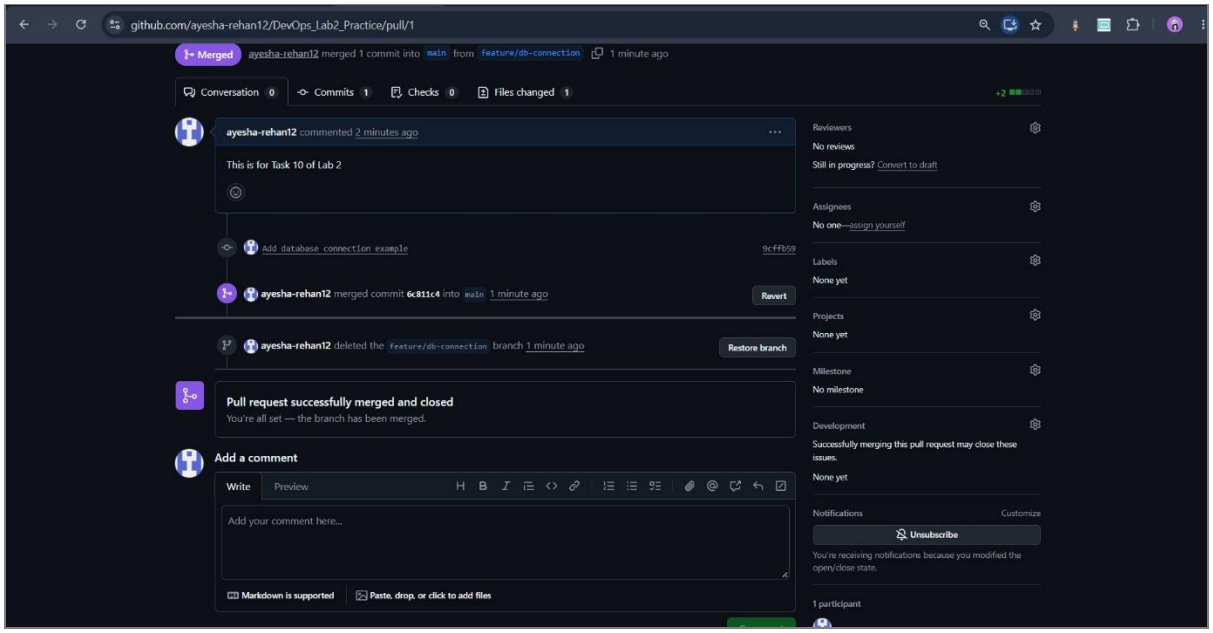


Figure: feature/db-connection branch deleted after merge.

Task 11: Detailed Branch Strategy (Develop/Staging)

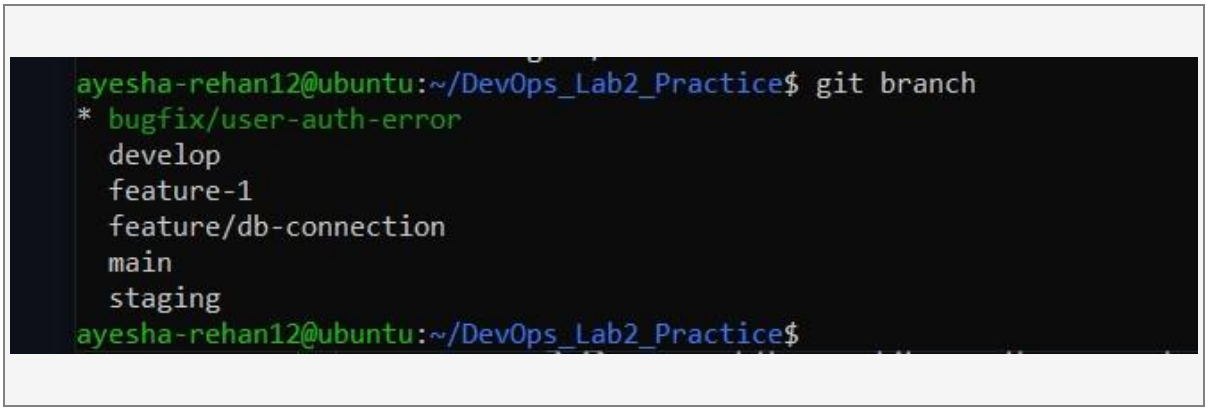


Figure: Branch list showing develop, staging, feature/ and bugfix/* branches.*

```
   6c811c4..848e151  develop -> develop
ayesha-rehan12@ubuntu:~/DevOps_Lab2_Practice$ git status
On branch develop
Your branch is up to date with 'origin/develop'.

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        __pycache__/
        error

nothing added to commit but untracked files present (use "git add" to track)
ayesha-rehan12@ubuntu:~/DevOps_Lab2_Practice$ git checkout staging
Switched to branch 'staging'
Your branch is up to date with 'origin/staging'.
ayesha-rehan12@ubuntu:~/DevOps_Lab2_Practice$ git pull --no-rebase origin staging
From github.com:ayesha-rehan12/DevOps_Lab2_Practice
 * branch            staging    -> FETCH_HEAD
Already up to date.
ayesha-rehan12@ubuntu:~/DevOps_Lab2_Practice$ git merge develop
Updating 6c811c4..848e151
Fast-forward
 feature1_notes.txt | 2 ++
 1 file changed, 2 insertions(+)
 create mode 100644 feature1_notes.txt
ayesha-rehan12@ubuntu:~/DevOps_Lab2_Practice$ git push origin staging
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:ayesha-rehan12/DevOps_Lab2_Practice.git
   6c811c4..848e151  staging -> staging
ayesha-rehan12@ubuntu:~/DevOps_Lab2_Practice$
```

Figure: Merges of feature/bugfix branches into develop and develop into staging.

```
ayesha-rehan12@ubuntu:~/DevOps_Lab2_Practice$ git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
ayesha-rehan12@ubuntu:~/DevOps_Lab2_Practice$ git pull --no-rebase origin main
From github.com:ayesha-rehan12/DevOps_Lab2_Practice
 * branch            main       -> FETCH_HEAD
Already up to date.
ayesha-rehan12@ubuntu:~/DevOps_Lab2_Practice$ git merge staging
Updating 6c811c4..848e151
Fast-forward
 feature1_notes.txt | 2 ++
 1 file changed, 2 insertions(+)
 create mode 100644 feature1_notes.txt
ayesha-rehan12@ubuntu:~/DevOps_Lab2_Practice$ git push origin main
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:ayesha-rehan12/DevOps_Lab2_Practice.git
   6c811c4..848e151  main -> main
ayesha-rehan12@ubuntu:~/DevOps_Lab2_Practice$ git status
On branch main
Your branch is up to date with 'origin/main'.

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        __pycache__/
        error

nothing added to commit but untracked files present (use "git add" to track)
ayesha-rehan12@ubuntu:~/DevOps_Lab2_Practice$
```

Figure: Final merge of staging into main.

Task 12: Code Review Workflow

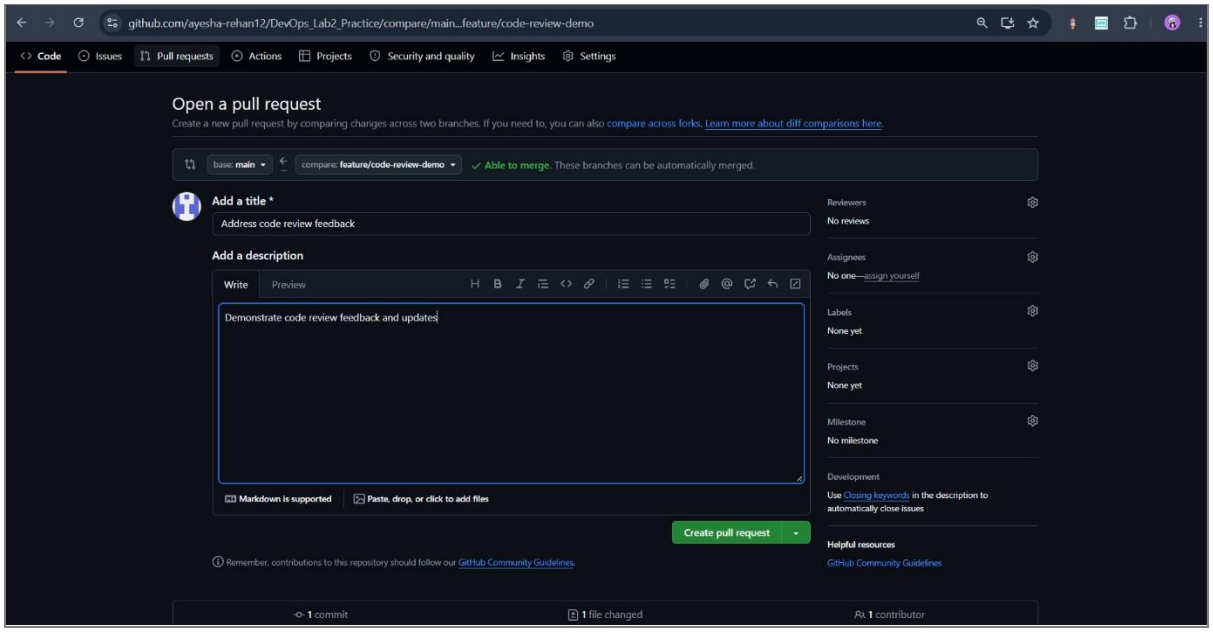


Figure: Pull Request with clear title and description.

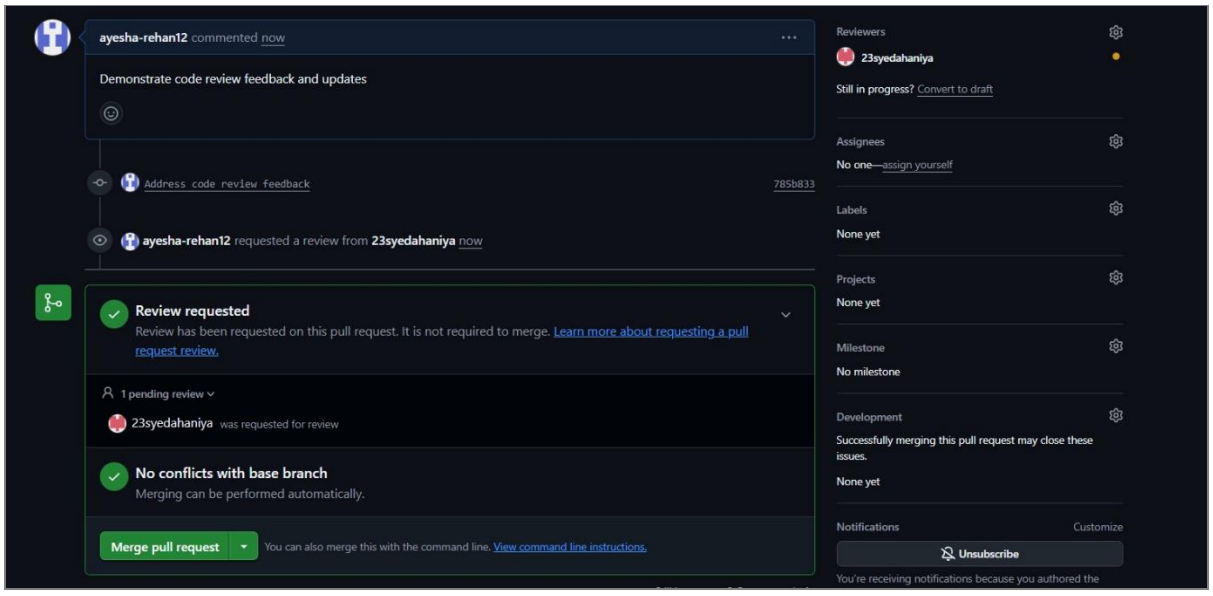


Figure: Reviewer assigned to the Pull Request.

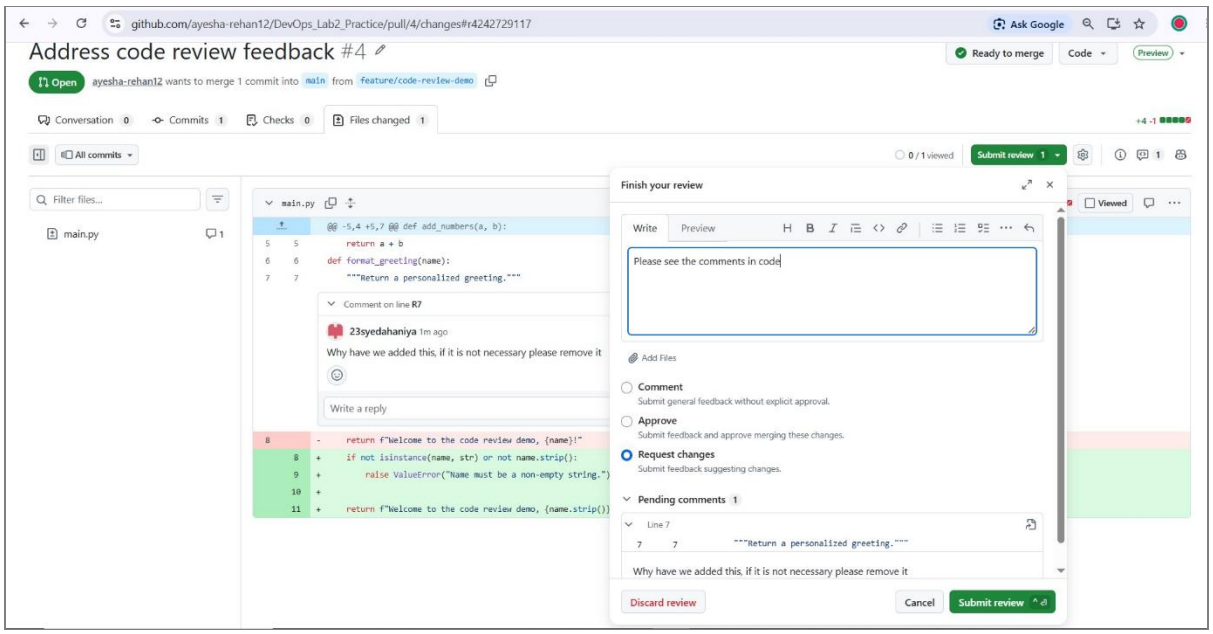


Figure: Reviewer requesting changes with comments.

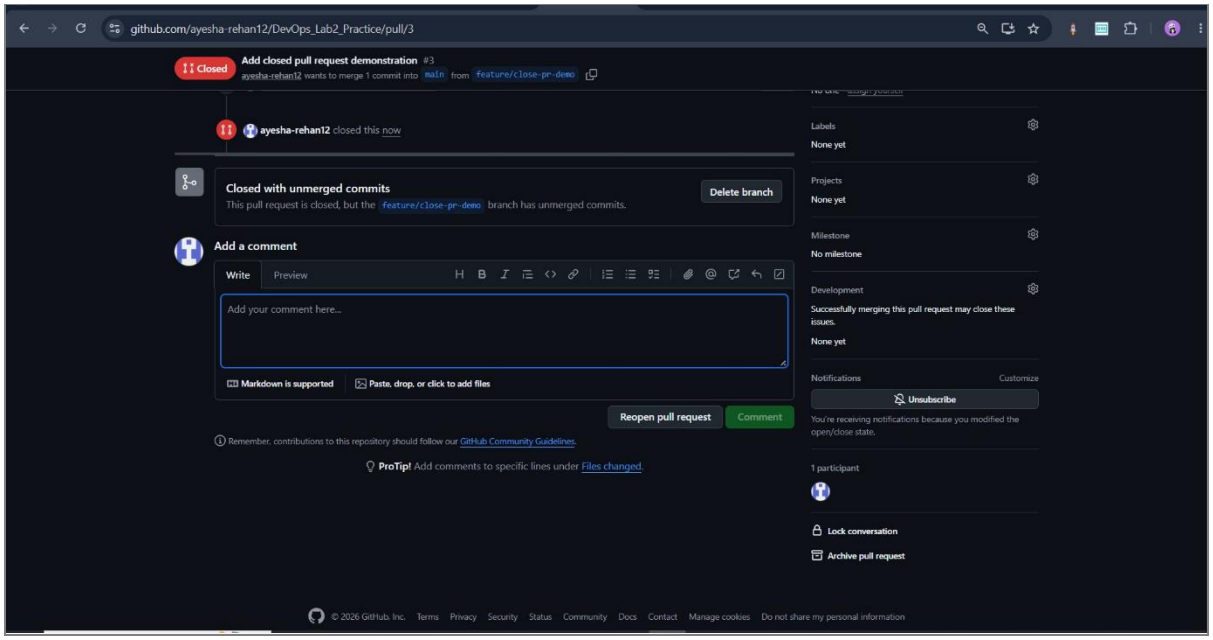


Figure: Practice Pull Request rejected/closed.

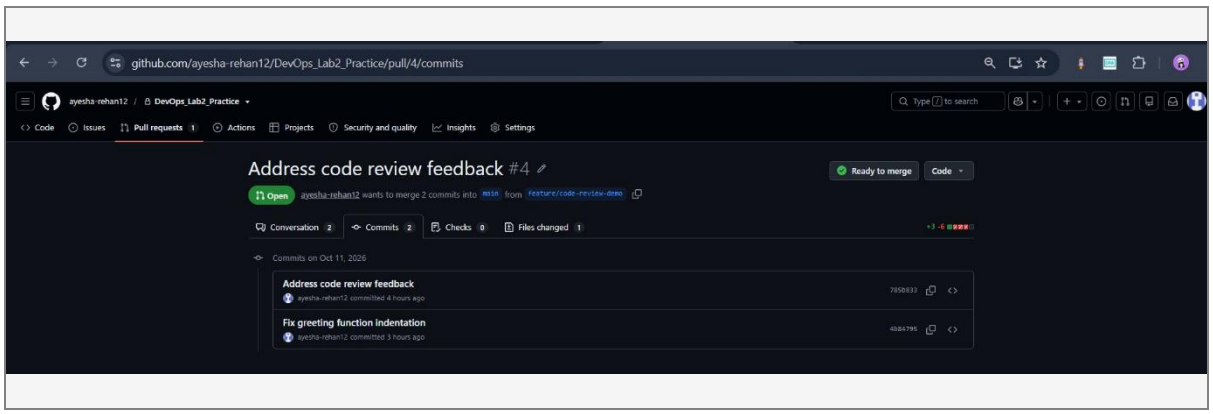


Figure: Pull Request automatically updated with new commits.

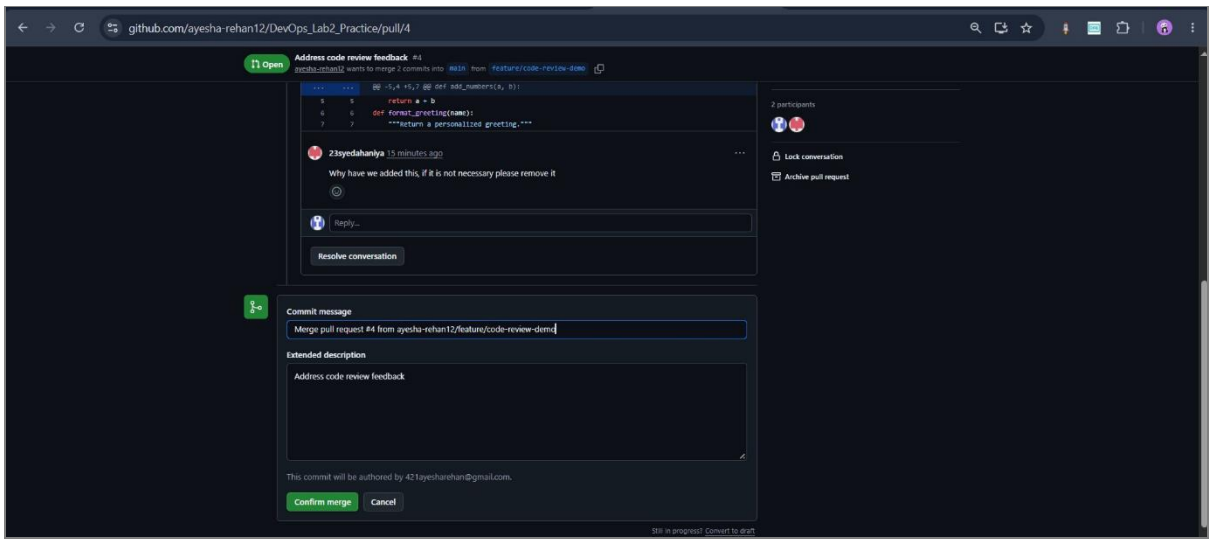


Figure: Merge confirmation after reviewer approval.

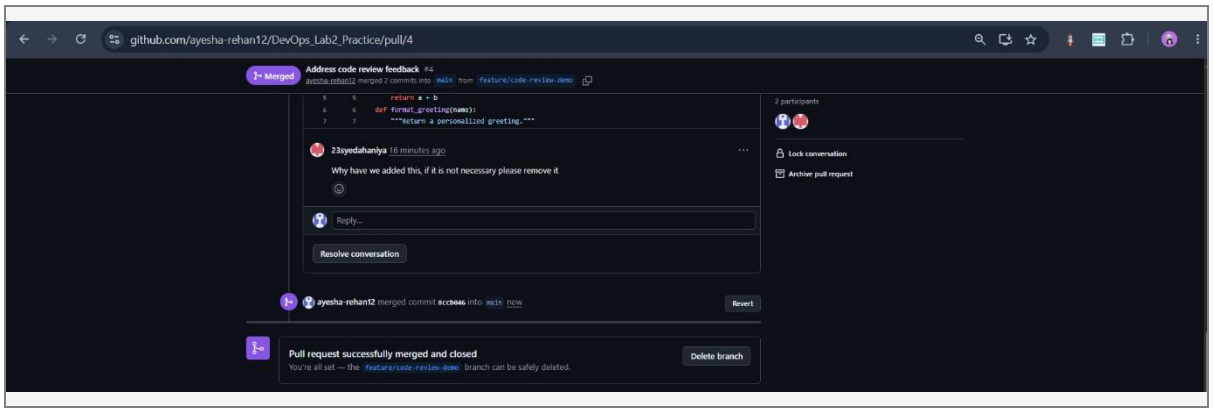


Figure: Pull Request successfully merged.

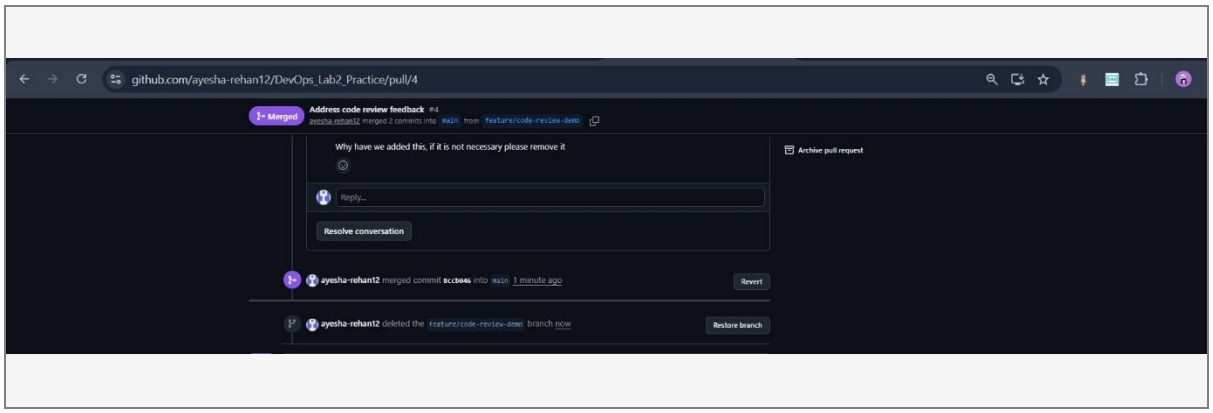


Figure: Merged PR branch deleted.

Task 13: Branch Cleanup Best Practices

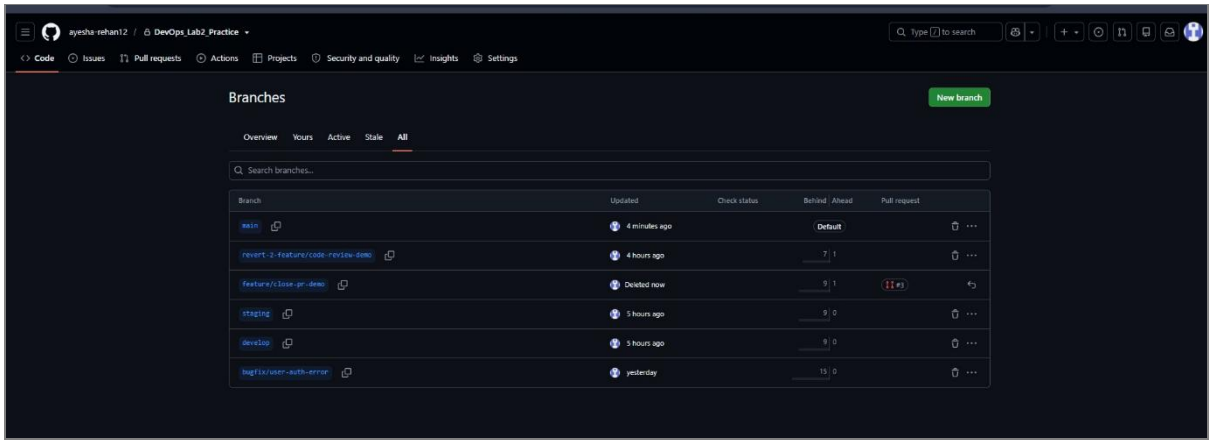


Figure: Merged remote branch deleted through the GitHub UI.

```
ayesha-rehan12@ubuntu:~/DevOps_Lab2_Practice$ git push origin --delete feature-1
To github.com:ayesha-rehan12/DevOps_Lab2_Practice.git
 - [deleted]         feature-1
ayesha-rehan12@ubuntu:~/DevOps_Lab2_Practice$ git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
ayesha-rehan12@ubuntu:~/DevOps_Lab2_Practice$ git pull origin main
remote: Enumerating objects: 2, done.
remote: Counting objects: 100% (2/2), done.
remote: Compressing objects: 100% (2/2), done.
remote: Total 2 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (2/2), 1.78 KiB | 606.00 KiB/s, done.
From github.com:ayesha-rehan12/DevOps_Lab2_Practice
 * branch            main       -> FETCH_HEAD
   99a9e87..49a89b5  main       -> origin/main
Updating 99a9e87..49a89b5
Fast-forward
 main.py | 9 +++------
 1 file changed, 3 insertions(+), 6 deletions(-)
ayesha-rehan12@ubuntu:~/DevOps_Lab2_Practice$ git branch -d feature-1
Deleted branch feature-1 (was da60a20).
ayesha-rehan12@ubuntu:~/DevOps_Lab2_Practice$ git branch
  bugfix/user-auth-error
  develop
  feature/close-pr-demo
  feature/code-review-demo
  feature/code-review-demo-final
  feature/code-review-update-demo
  feature/db-connection
* main
  staging
ayesha-rehan12@ubuntu:~/DevOps_Lab2_Practice$
```

Figure: Remote branch deleted from terminal using git push origin --delete.

Bonus Task: Simulated Team Collaboration

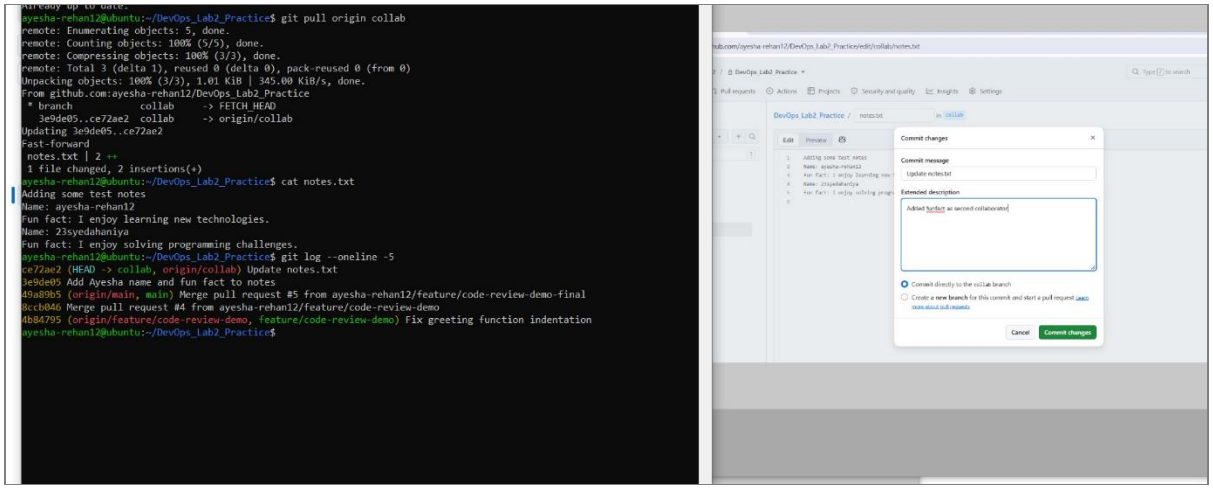


Figure: Collaborators' commits (name and fun fact) pushed to the collab branch.

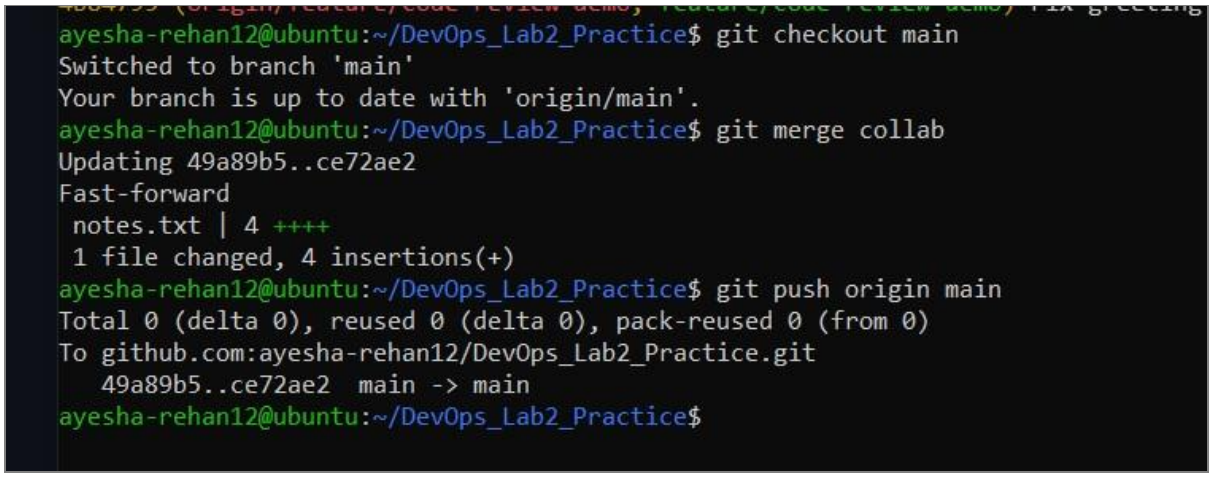


Figure: collab branch merged into main and pushed.

Exam Question 1: Advanced Branching & Merge Verification



Figure: New branch created.

```
ayesha-rehan12@ubuntu:~/DevOps_Lab2_Practice$ git checkout -b exam-q1-branch
Switched to a new branch 'exam-q1-branch'
ayesha-rehan12@ubuntu:~/DevOps_Lab2_Practice$ git add notes.txt
ayesha-rehan12@ubuntu:~/DevOps_Lab2_Practice$ git commit -m "Complete Q1 branch change"
On branch exam-q1-branch
Untracked files:
  (use "git add <file>..." to include in what will be committed)
        __pycache__/
        error

nothing added to commit but untracked files present (use "git add" to track)
ayesha-rehan12@ubuntu:~/DevOps_Lab2_Practice$
```

Figure: Small change made and committed.

```
ayesha-rehan12@ubuntu:~/DevOps_Lab2_Practice$ git merge exam-q1-branch
Already up to date.
```

Figure: Branch merged back into main.

```
ayesha-rehan12@ubuntu:~/DevOps_Lab2_Practice$ git log --oneline --graph --all
* ce72ae2 (HEAD -> exam-q1-branch, origin/main, origin/collab, main, collab) Update notes.txt
* 3e9de05 Add Ayesha name and fun fact to notes
*   49a89b5 Merge pull request #5 from ayesha-rehan12/feature/code-review-demo-final
|\
| * f5dd862 (origin/feature/code-review-demo-final, feature/code-review-demo-final) Fix greeting function indentation
| * 334bd79 Address code review feedback
| * 865e760 Add reviewed greeting validation
* |   8ccb046 Merge pull request #4 from ayesha-rehan12/feature/code-review-demo
|\ \
| |/
|/|
| * 4b84795 (origin/feature/code-review-demo, feature/code-review-demo) Fix greeting function indentation
| * 785b833 Address code review feedback
* | 99a9e87 (feature/code-review-update-demo) Merge pull request #2 from ayesha-rehan12/feature/code-review-demo
|\|
| * 4c48f1a Add code review demonstration
|/
| * 6f4266a (origin/feature/close-pr-demo, feature/close-pr-demo) Add closed pull request demonstration
|/
*   848e151 (origin/staging, origin/develop, staging, develop) Merge branch 'develop' of github.com:ayesha-rehan12/DevOps_Lab2_Practice into
|\
| *   6c811c4 Merge pull request #1 from ayesha-rehan12/feature/db-connection
| |\
| | * 9cffb59 (origin/feature/db-connection, feature/db-connection) Add database connection example
* | | da60a20 Add feature 1 notes
|/ /
* / 58f172a Add new function to main.py
|/
* 8ea7ef7 Add notes.txt
:...skipping...
* ce72ae2 (HEAD -> exam-q1-branch, origin/main, origin/collab, main, collab) Update notes.txt
* 3e9de05 Add Ayesha name and fun fact to notes
*   49a89b5 Merge pull request #5 from ayesha-rehan12/feature/code-review-demo-final
|\
| * f5dd862 (origin/feature/code-review-demo-final, feature/code-review-demo-final) Fix greeting function indentation
| * 334bd79 Address code review feedback
| * 865e760 Add reviewed greeting validation
* |   8ccb046 Merge pull request #4 from ayesha-rehan12/feature/code-review-demo
|\ \
| |/
```

Figure: Commit history verifying the merge.